The Bootstrap: Recap

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Sampling from $\hat{F}$

We need the sampling distribution of $T = T(X_1, \ldots, X_n)$, $X_i \overset{\text{iid}}{\sim} F$, and F is unknown.

- Replace F with the **empirical CDF**, $\hat{F}(t) = \frac{1}{n}\sum_i \mathbf{1}(X_i \le t)$, which puts mass $1/n$ on each observation.
- Inverting $\hat{F}$ at $U \sim U(0,1)$ picks one observation uniformly at random, so n draws from $\hat{F}$ is `sample(x, replace = TRUE)`.

The bootstrap. For $b = 1, \ldots, B$, draw a bootstrap sample $X_1^*, \ldots, X_n^*$ and compute T_b^*. The SD of $T_1^*, \ldots, T_B^*$ is the **bootstrap SE**.

- We draw B of the n^n possible bootstrap samples. Larger B reduces Monte Carlo error, but not the error from using $\hat{F}$ for F.
- Check on $\bar{X}$ first: its bootstrap SE should be close to $s/\sqrt{n}$.

Three intervals

With T^*_p the p quantile of the bootstrap distribution:

interval	formula	idea
normal	$T \pm z_{1-\alpha/2}\,\widehat{\mathsf{SE}}$	T is approximately normal
basic	$[2T - T^*_{1-\alpha/2},\ 2T - T^*_{\alpha/2}]$	$T^* - T$ behaves like $T - \theta$
percentile	$[T^*_{\alpha/2},\ T^*_{1-\alpha/2}]$	$g(T) - g(\theta)$ is symmetric for some increasing g

- The basic and percentile intervals are reflections of each other about T. They agree when the bootstrap distribution is symmetric about T.
- Better intervals exist. The bootstrap-t comes on day 2, with the jackknife.

Flint: the 90th percentile

The Lead and Copper Rule compares the 90th percentile with 15 ppb. It has no simple SE formula.

MDEQ samples	T	SE	normal	basic	percentile
all 71	18	5.9	[6.4, 29.6]	[7.0, 28.0]	[8.0, 29.0]
69, without two	11.4	4.8	[2.1, 20.7]	[0.0, 15.6]	[7.2, 22.8]

- Excluding two samples moves T by about one SE. Every interval contains 15.
- Without them the bootstrap distribution is skewed, and the intervals disagree.
- The bootstrap treats the homes as a random sample. It cannot detect how they were chosen.

Paired data

Virginia Tech took two bottles from each home: the first draw, and one after 2 minutes of flushing.

- $\hat{F}$ puts mass $1/n$ on each **pair**, so resample **rows**: `i <- sample(n, replace = TRUE)`, then use `x[i]` and `y[i]`.
- Resampling each column separately draws X^* and Y^* independently. The bootstrap correlations center on zero, and the SE of the mean difference of logs grows from 0.07 to 0.12.

Failure case: the sample maximum

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$ and $T = X_{(n)}$. A bootstrap sample can only return observed values, so

$$P(X^*_{(n)} = x_{(n)}) = 1 - \left(1 - \frac{1}{n}\right)^n \to 1 - e^{-1} \approx 0.632,$$

while the true distribution of $X_{(n)}$ is continuous.

- $T^* \leq x_{(n)} < \theta$, so the percentile interval **never** covers θ. The basic and normal intervals undercover too.
- The bootstrap needs a small change in F to make a small change in T: smooth functions of means work, quantiles roughly, the max and min not at all.
- A spike at T in the bootstrap distribution is a warning sign.